11 Ch 11 – International Financial Management

Asked in last 3 papers: Sep 2025 Q5(a) 7m foreign project NPV (forward rates from inflation); Jan 2026 Q5(b) 6m GDR number and cost; May 2026 Q1(a) 6m foreign project with swap and forward rates; May 2026 Q3(a) 6m forward rates by IRP + APV.

1. International Capital Budgeting · 1.1 Complexities Involved

- → *Foreign cash flows must be **converted into parent's currency**.*
- → ***Parent cash flows differ** from project cash flows.*
- → *Remitted profits taxed in **home and host** country.*
- → ***Foreign exchange risk** on parent's cash flow.*
- → ***Inflation changes** shift the competitive environment and cash flows.*
- → *Host-country **restrictions on distribution** of cash flows.*
- → *Initial investment in host country to benefit from **release of blocked funds**.*
- → ***Political risk** reduces chance of expected cash flows.*
- → *Host-country **concessions/benefits** raise profitability.*
- → ***Terminal value** hard to estimate — buyers in parent company have divergent views.*

1.2 Problems Affecting Foreign Investment Analysis

- → ***Foreign exchange risk:** forecast host-country inflation over project life; adjust local-currency cash flows for inflation; convert at **spot rate × expected depreciation from PPP**.*
- → ***Remittance restrictions:** limits on transfer of profits, depreciation, technical differences → project CF ≠ parent CF. Diluted by **internal transfer prices, overhead payments**. Blocked funds adjusted at their **opportunity cost**.*
- → ***Taxation:** use **after-tax** cash flows. Two tax regimes, royalties/dividends/management fees, withholding tax, tax treaties, tax discrimination (remitted vs reinvested profits) complicate it. MNCs cut overall tax via **transfer pricing**.*
- → *Rule: compute parent's after-tax CF at the **higher of home/host tax rate**. Feasible → acceptable under a more favourable regime too. Not feasible → bring in other tax-saving aspects to see if it crosses the hurdle rate.*

1.3 Project vis-a-vis Parent Cash Flows

- → *Difference arises from **tax rules, exchange controls**; management fees and royalty are returns to the parent.*
- → ***Project (own) CF basis:** project must beat local competitors, else parent would buy local equity/government bonds. But foreign projects often **replace imports**, not compete locally. Avoids currency conversion and exchange-rate problems.*
- → ***Parent CF basis:** operating + financial cash flows **actually remitted** are the yardstick (dividends to shareholders, debt service). Evaluate on **net after-tax operating CF**; keep financial CF separate from operating CF.*

1.4 Discount Rate and Adjusting Cash Flows

- → *Old practice: raise discount rate for foreign risk. **Not proper** to combine all risks into one rate:*
  - → *Political-risk premium in the rate **penalises early cash flows more** than distant ones.*
  - → *Raising the rate for adverse currency moves ignores **gains from favourable moves**.*
- → *Better: **adjust cash flows** (certainty equivalents, probability, decision tree) for political, exchange and other risks; discount at host- or parent-country rate.*

Forward / expected rates and the two approaches (needed for 1.6 and the TYK practicals):

Formula	Terms / logic	SM example figures
PPP: $E(S_1) = S_0 \times (1 + I_d) \div (1 + I_f)$	S_0 = spot (**direct quote**, home per 1 foreign); I_d = home inflation; I_f = foreign inflation. Higher-inflation currency depreciates. Year n: roll forward year by year (or use power n if rates constant).	Illus 1: ₹7 × 1.06/1.05 = ₹7.54
IRP: $F_n = S_0 \times (1 + i_d)^n \div (1 + i_f)^n$	Same logic with nominal interest rates ("international parity conditions hold").	Illus 3: 9.50 × 1.12/1.10 = 9.67; year 2 = 9.85
Indirect quote (foreign per 1 home): $F = S \times (1 + rate_{foreign}) \div (1 + rate_{home})$	Numerator is always the currency in which the quote is expressed. Then ₹ CF = NC CF ÷ rate.	TYK PS: NC 1.60 × 1.09/1.08 = 1.615, 1.630, 1.645
Home currency approach: foreign CF × forward rate → discount at home rate	Convert year by year at expected rates, discount at the home (₹) rate.	Illus 3(b), TYK P3, TYK PS
Foreign currency approach: $(1 + k_{home}) = (1 + i_{home})(1 + RP)$; $k_{foreign} = (1 + RP)(1 + i_{foreign}) - 1$	RP = project risk premium. Discount foreign CF at the foreign risk-adjusted rate; foreign NPV × spot = home NPV.	TYK P1: 1.14/1.12 = 1.0179; × 1.08 → 9.9%

Linking line (app-written): both approaches give the same answer when parity holds; examiners say which one to use ("foreign currency approach" in TYK P1; "converted at forward rate and discounted at the Indian rate" in TYK PS).

1.5 Adjusted Present Value (APV)

Asked: May 2026 Q3(a) 6m

→ **Value-additive** approach: each cash flow discounted separately at a rate matching **its own risk**.

→ Test **basic viability** first; if feasible, stop. If not, add the other complexities (tax shields, subsidies).

Formula	Terms / logic
$APV = -I_0 + \Sigma\, X_t \div (1 + k^*)^t + \Sigma\, T_t \div (1 + i_d)^t + \Sigma\, S_t \div (1 + i_d)^t$	I_0 = PV of investment outlay (net of usable blocked funds); X_t = operating CF, discounted at **k^* = unlevered cost of capital**; T_t = tax saving from financing mix (interest tax shield); S_t = before-tax value of interest subsidies from project-specific financing; last two at **i_d = before-tax cost of debt** (they are as certain as debt).

→ **Blocked funds:** balances abroad that cannot be remitted. **Direct** = exchange-control bar. **Indirect** = heavy local/withholding tax on remittance. If released for the new project, they **reduce the investment outlay**.

1.6 Scenarios · 1.6.1 A foreign company investing in India

Asked: Sep 2025 Q5(a) 7m; May 2026 Q1(a) 6m

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Illus 1 (Perfect Inc., US, JV in Baddi): 55% revenue share + royalty $0.01/bottle; outlay ₹200 cr, scrap ₹5 cr; depreciation 20% of 195 = 39.00; WC 50 + 5 p.a.; 24 cr bottles at ₹7.50 (+₹1 each year, plus that year's inflation); production cost 40%. Tax India 30%, US 35% with credit, paid next year. Spot ₹57; inflation India 6% (falls 0.50% p.a.), US 5%. Only **50% remittable** in the year, rest next year. Rate 15%.

PPP rates yr 0–6: 57, 57.54, 57.82, 57.82, 57.54, 56.99, 56.18. Inflated prices 7.95, 8.97, 9.98, 10.97, 11.96 → 55% share 104.94 … 157.87. Royalty ₹13.81 … 13.68 cr. PBT 37.77 … 69.40; Indian tax 11.33 … 20.82. FCF: −250.00, 71.77, 68.59, 74.15, 79.51, 164.89, −20.82. Remittance (50% + previous 50%): −250.00, 35.88, 70.17, 71.37, 76.84, 122.20, 61.62 → in $ mn −43.86, 6.24, 12.14, 12.34, 13.35, 21.44, 10.97. US tax 35% (next year) less Indian tax credit → net tax; net CF −43.86, 6.24, 11.92, 10.47, 11.84, 20.02, 7.18, −3.84. **NPV @15% = −$4.15 mn → do not invest.**

Logic: compute project CF in host currency → apply remittance restriction → convert at PPP rates → home tax less foreign tax credit (with the one-year lag) → discount at the parent's required rate.

1.6.2 Indian company investing abroad, funds raised in the same country

→ Use the **foreign competitor's beta** (business risk of that market): ungear, then regear at the project's own funding mix, with host-country tax.

Formula	Terms / logic
$\beta_A = \beta_E \times E \div [E + D(1 - t)]$	Debt assumed risk-free ($\beta_D = 0$). Same formula reversed to get project β_E.
$K_e = R_f + \beta(R_m - R_f)$; $WACC = K_e \cdot W_e + K_d(1 - t) \cdot W_d$	CAPM; host-country tax rate on debt.

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Illus 2 (Its Entertainment Ltd., Water Park in Nepal): competitor E 1850, D 510, β_E 1.35, Nepal tax 20% → β_A **1.106**; regeared at 55:45 → β_E **1.83**; K_e = 8% + 1.83(11% − 8%) = **13.49%**; WACC = 13.49% × 0.55 + 9%(1 − 0.20) × 0.45 = **10.66%**. Construction NPR 450 cr (225 now, 225 end yr 1); WC 65 cr from yr 1, rising 5%; capital allowance 20% WDV on max NPR 200 cr (40, 32, 25.60, 20.48). Visitors 20,000 × 70% × 360 = 50,40,000; revenue per visitor 303.19 (ticket 242.55 + F&B profit 33.08 + gifts 27.56) in first operating year. Market research ₹ 20 lakh (sunk) and apportioned overheads NPR 5 cr excluded. Net CF: −225.00, −290.00, 45.26, 43.32, 41.84, 369.79 → **NPV = NPR −167.32 cr → do not undertake.**

1.6.3 Indian company investing abroad, funds raised in a different country through GDRs

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Illus 3 (Opus Technologies, project in China, 2 years): cost of GDR $K_e = D_1 \div P_{net} + g$ = 2.75 ÷ 198 + 0.10 = **11.39%** (D_0 25% of FV ₹ 10 = 2.50; net proceeds ₹ 200 − 1%). Land CN¥ 30,00,000 sold for 35,00,000 → net of 25% tax 33,75,000; office complex 15,00,000 (SLM to zero) sold for 5,00,000 → net 3,75,000. Operating CF 8,35,500 and 8,87,340. (a) **NPV in CN¥ @11.39% = −12,025**. IRP rates 9.67, 9.85. (b)(i) remitted each year, discount @12%: **NPV ₹ 8,70,008**; (ii) all remitted at end of yr 2 (CN¥ 54,72,840): **NPV ₹ 2,14,257**. CN¥ NPV negative but ₹ NPV positive due to weakening of ₹ against CN¥ → **accept**.

Logic of (b)(ii): remittance restriction delays the inflow, so PV falls (8,70,008 → 2,14,257).

TYK practicals on international capital budgeting (1.1–1.6):

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TYK P1 (ABC Ltd., US project, foreign currency approach): outlay $1,10,00,000; inflows $20, 25, 30, 40, 50 lakh; spot ₹ 48; R_f US 8%, India 12%; required 14%. 1 + RP = 1.14/1.12 = 1.0179 → $ rate = 1.0179 × 1.08 − 1 = **9.9%**; PV $12.013 mn − 11.000 = **$1.013 mn** → **₹ 48.624 mn**.

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TYK P2 (US company, software unit in India, transfer price $10 mn): revenue ₹ 48,00,00,000; costs rent 15,00,000 + manpower 400 × 80 × 10 × 365 = 11,68,00,000 + admin 12,00,000 = 11,95,00,000; EBT 36,05,00,000; tax 30% 10,81,50,000; EAT 25,23,50,000.

(i) WHT 10% 2,52,35,000 → repatriation ₹ 22,71,15,000 = **$4.7 mn**; cost to US company **$5.268 mn** vs sale $12.0 mn → develop in India.

(ii) Alt (no WHT as subsidiary paid tax): repatriation 25,23,50,000 = **$5,257,292**; cost **$4.743 mn**.

(iii) Alt (WHT 10% on EBT 3,60,50,000, credited against corporate tax 7,21,00,000): repatriation same 25,23,50,000 = $5,257,292; cost $4.743 mn.

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TYK P3 (XY Ltd., stores in Africa, real CF): inflation Africa 40%, India 10%; spot Rand 6/₹; rate 20%. Rates 6.00, 7.6364, 9.7190, 12.3696. Nominal ₹ CF −50,000, −1,650, −2,420, −3,327.50; Rand nominal −2,00,000, 70,000, 1,37,200, 2,46,960 → ₹ −33,333, 9,167, 14,117, 19,965. Net −83,333, 7,517, 11,697, 16,637; PV 3 years = **−59,320**; terminal value 16,637 ÷ 0.20 × 0.579 = **48,164**; **NPV = −11,156 (₹ '000)**.

Note (app-written): ICAI labels the rates "as per IRP", but they are computed from the inflation rates (PPP). Figures kept as printed.

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TYK P4 (MNC subsidiary in India, replacing exports): outflow 500 + 50 − release of existing WC 15 = **$535 mn**. CFAT India: sales 5 mn × $80 = 400 − VC 100 − cash fixed 30 − depreciation 100 = EBIT 170; tax 35% 59.50; CFAT **210.50**. Allocated fixed cost $3 mn ignored (not incremental). Lost export CFAT: 1.5 mn × (80 − 40) = 60 − tax 21 = **39.00**. Incremental CFAT 171.50 × 3.6048 = 618.2232 + WC release 35 × 0.5674 = 19.8590 → 638.0822 − 535 = **NPV $103.0822 mn → accept**.

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TYK P5 (XY2 Ltd., import of wood from Nepal; same type as Sep 2025 Q5(a)): NC flows −25.000, 2.600, 3.800, 4.100; ₹ flows 0, 2.869, 4.200, 4.600; rate 9%; inflation Nepal 9%, India 8%; ₹ 1 = NC 1.6. Forward NC/₹ 1.615, 1.630, 1.645 → ₹ −15.625, 1.61, 2.33, 2.49. Total 4.479, 6.53, 7.09 → **NPV ₹ −0.547 mn**. MIRR: terminal value 5.32 + 7.12 + 7.09 = 19.53; MIRR = $(19.53 \div 15.625)^{1/3} - 1$ = **7.72%**.

MIRR = (Terminal value of inflows reinvested at k ÷ Initial outlay)$^{1/n}$ − 1. TV uses the FV factors given (1.188 for 2 years, 1.090 for 1 year).

2. International Sources of Finance · 2.1 Foreign Currency Convertible Bonds (FCCBs)

- → *Convertible bond issued in a **currency other than issuer's domestic currency**; mix of debt and equity; regular coupon + principal, plus **option to convert** into stock.*
- → *Investors: safety of guaranteed payments + share in price appreciation (via warrants activated at a set price). Issuer: **lower coupon** due to equity element.*

Advantages	Disadvantages
→ ***Flexibility** to convert into equity or redeem at end of period (normally 3 years).*	→ ***Exchange risk** on interest in foreign currency → only firms with low D/E and large forex earnings opt.*
→ ***Delayed dilution** of equity and EPS.*	→ ***More debt** and forex outgo on interest.*
→ ***Easily marketable**: conversion option + minimum fixed interest.*	→ *Low interest (around 3–4%) but exchange risk on interest and repayment if not converted. Only major plus: high-growth firm can issue shares later above current price.*

2.2 American Depository Receipts (ADRs)

- → *Depository receipts issued in the **USA**; must follow stringent **SEC** provisions.*
- → *Created by deposit of non-US company's securities with a **custodian bank in issuer's country**; custodian informs the US depository, which issues ADRs.*
- → ***USD denominated**, traded like US securities; holder gets same rights as owners of underlying shares.*
- → *GDR = same structure, but traded in **more than one currency** and within as well as outside the US.*

2.3 Global Depository Receipts (GDRs)

Asked: Jan 2026 Q5(b) 6m

- → **Negotiable certificate**, denominated in a currency not native to the issuer, representing its publicly traded local-currency equity shares. Mostly USD, a few in Euro/Pound. Issued in US = ADR; outside US = GDR. Can represent debt in theory, rarely in practice.
- → Created when local shares are delivered to the depository's **local custodian bank**; depository bank (e.g. Bank of New York) issues DRs in USD; traded on a foreign exchange, OTC, or among **QIBs**. Indian issues are GDRs as they are marketed globally.
- → Why Indian companies used them: equity (no principal repayment, lower servicing cost); no other form of term forex funding was available; fetched **higher prices** than a domestic issue.

Impact of GDRs on Indian capital market (TYK T2):

- → Market shifting from **Bombay to Luxemburg**.
- → **Arbitrage** possibility in GDR issues.
- → Market **no longer independent** of the world — investors must track worldwide events.
- → **Retail investors sidelined** — GDR/FII placements + free pricing end easy money on discounted rights/public issues.
- → Result: considerable foreign investment flowed into India.

Markets of GDRs:

- → Sold primarily to **institutional investors**; demand dominated by **emerging market funds**.
- → FIIs likely to **switch** from ordinary shares into GDRs.
- → Major demand: **UK, USA (QIBs), South East Asia** (Hong Kong, Singapore), some continental Europe (France, Switzerland).

Mechanism: company issues ordinary shares → kept with custodian/depository bank → GDRs issued to foreign investors.

Characteristics:

- → Economic benefits of shareholders but **no voting rights**.
- → Settled through **CEDEL & Euro-clear**; listed on **Luxemburg** stock exchange.
- → Traded **OTC** between professional market makers; **freely traded**; marketed globally in more than one currency.
- → Dividends paid in issuer currency, converted into dollars by depository → **exchange risk with investor**.
- → **Cancellation** any time after a cooling period of **45 days**: depository asks domestic custodian to release underlying shares; BSE/NSE price on date of redemption advice = cost of acquisition.

Formula	Terms / logic	SM example figures
Gross issue = Net requirement ÷ (1 − flotation %)	Funds needed are net of issue cost.	Illus 4: $10 mn ÷ 0.98 = $10.204 mn
Price per GDR = MP × shares per GDR × (1 − discount) ÷ exchange rate	Converted to $ at expected rate.	250 × 2 × 90% = ₹ 450 ÷ 60 = $7.50
No. of GDRs = Gross issue ÷ price per GDR		10.204 ÷ 7.50 = **1.3605 mn**
Cost of GDR K_e = D_1 ÷ Net proceeds per GDR + g	D_1 and net proceeds both per GDR (in ₹).	40 ÷ 441 + 0.12 = **21.07%**

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Illus 4 (X Ltd., plant in US): need $10 mn net, flotation 2%; MP ₹ 250 (FV ₹ 100); 2 shares per GDR at 10% discount; ₹ 60/$; dividend 20% (₹ 20 per share → D_1 ₹ 40 per GDR), growth 12%. Net proceeds per GDR ₹ 450 × 0.98 = ₹ 441.00. (a) **1.3605 million GDRs**; (b) **cost 21.07%**. Also see Illus 3: 2.75 ÷ 198 + 0.10 = 11.39%.

2.4 Euro-Convertible Bonds (ECBs)

→ Debt with option to convert into a **predetermined number of equity shares**; conversion price usually at a **premium**; **fixed interest**.

→ **Call option (issuer's):** issuer can call bonds before maturity when share price is far above redemption value — **forces conversion**; usually when price reaches **130% to 150%** of conversion price.

→ **Put option (holder's):** holder can sell bonds back to issuer at a pre-determined price and date.

→ Interest and redemption paid by issuer in **US dollars**.

2.5 Other Sources

→ **Euro-convertible zero coupon bonds:** no interest; converted at maturity at a pre-determined price; usually 5 years; a **deferred equity issue**.

→ **Euro-bonds with equity warrants:** market-based coupon; **detachable** warrants; pure bonds trade at a discount; suit fixed income funds.

→ **Syndicated bank loans:** large loans from banks, good credit rating, quick, few formalities; 5 to 10 years; rate = index (say **SOFR**) + spread by credit rating; covenants (key ratios).

→ **Euro-bonds:** debt denominated in a currency, issued **outside the country of that currency** (e.g. Yen bond floated in France); refuge from tax/regulation, yield arbitrage; usually bearer; forms: fixed rate, FRNs, convertible.

→ **Foreign bonds:** denominated in a currency foreign to the borrower and **sold in the country of that currency**; subject to that country's restrictions on foreigners.

→ **Euro commercial papers:** short-term, issued at a **discount**, maturity < 1 year.

→ **Credit instruments:** used for foreign remittances; differ in **speed** of receipt; rate varies with loss of interest and risk → different exchange rates for different instruments.

TYK TI (short note on instruments of international finance) → answer from 2.1–2.5.

3. International Financial Centre (GIFT City)

→ IFC: financial centre catering to customers **outside their own jurisdiction**; hub for funds, products and services in own land but under a **different set of regulations and laws**; flexibility in currency trading, insurance, banking.

3.1 Benefits of IFC

→ Qualified professionals working abroad can **come and practise** here.

→ Platform to pursue **global opportunities without leaving homeland**.

→ Stops **brain drain**.

→ **Brings back** financial services transactions now done abroad.

→ Trading of **complicated financial derivatives** from India.

3.2 Constituents of IFC

→ **Highly developed infrastructure**.

→ **Stable political environment** (avoids country risk).

→ **Strategic location** — near airport, seaport; friendly weather.

→ **Quality life** — to retain highly paid professionals.

→ **Rational regulatory framework** — fair and transparent.

→ **Sustainable economy** — absorbs shocks, boosts investor confidence.

3.3 GIFT City — India's International Financial Services Centre

- Idea coined in **2007** to compete with Dubai, Hong Kong; aim: **retain financial services business** moving out of India; separate jurisdiction with relaxed tax and other laws.
- IFSC operationalised at **GIFT Multi Services SEZ in April 2015**; **Union Budget 2016** gave competitive tax regime.
- **India INX** (wholly owned subsidiary of BSE) inaugurated **9/1/2017** — index, currency, commodity, equity derivatives. **NSE** launched at GIFT on **3 June 2017**.
- Competitive cost and tax, **single window clearance**, relaxed company law, international arbitration centre; moving to **unified regulator**. **FEMA not applicable** at GIFT City. Units pay SEZ-level taxes and can offer foreign currency loans.

4. Sovereign Funds

- SWF: **state-owned investment fund** of money generated by government, generally from **surplus reserves**. Legal basis: constitutive law, fiscal law, constitution, company law or other laws.

Sources	Objectives	Classifications / vehicles
→ Surplus from state-owned **natural resource** revenues and **trade surpluses** → **Bank reserves** from budgeting excesses → **Foreign currency operations** → **Privatisation** money → **Governmental transfer payments**	→ Protect and **stabilise** budget/economy → **Diversify** from non-renewable exports → **Better returns** than forex reserves → Help monetary authorities dissipate **unwanted liquidity** → **Savings for future generations** → Social and economic **development** → Sustainable long-term capital growth for target countries → **Political strategy**	→ Classes: stabilisation; savings/future generation; public benefit pension reserve; reserve investment; strategic development SWF (SDSWF) → Vehicles: SWFs; public pension funds; state-owned enterprises; sovereign wealth enterprises (SWEs)

Risk appetite ranges from very conservative to high; some prefer returns over liquidity, some vice-versa.

5. International Working Capital Management

Reasons for complexity vs a domestic firm:

- **Wider financing options** — local or global markets.
- **Interest and tax rate differentials** between countries.
- **Foreign exchange risk** on inflows/outflows and imports/exports.
- Home/host **restrictions on movement of cash and inventory** (political).
- **Limited knowledge** of politico-economic conditions of host countries (eased by communication).
- **Capital convertibility:** full convertibility lets MNC position funds efficiently; not available in countries like India.

Covers multinational cash, inventory and receivables management.

S.1 Multinational Cash Management

- → Traditional objectives: (1) **manage and control cash** — better collections/disbursements, accurate forecasts; (2) **optimum utilisation and conservation** — cash when needed, minimum balance, higher risk-adjusted return.
- → Constraints: country rules, forex conventions, FDI restrictions, **repatriation time limits**, **blocked funds**, multiple tax jurisdictions and currencies, no integrated exchange facilities.

Objectives of effective international cash management:

- → Minimise **currency exposure** risk.
- → Minimise **overall cash requirement** without disturbing subsidiaries.
- → Minimise **transaction costs**.
- → Minimise **country's political risk**.
- → **Economies of scale** and superior knowledge.

Centralised cash management helps to:

- → Keep **minimum cash balance**; manage **liquidity** of the centre.
- → Use **hedging strategies** optimally to minimise exposure.
- → Generate **maximum returns** by investing all cash.
- → Use **multinational netting** — lower transaction costs and exposure.
- → Use **transfer pricing** to enhance profitability.
- → Exploit **currency correlations**: payables and receivables in positively correlated currencies; payables in negatively correlated currencies; **pooling** — group variance smaller than sum of individual variances.

Cash flows of an MNC: subsidiaries send fees, dividends, excess cash to parent; parent sends loans and return on cash invested; subsidiaries pay each other for supplies. Two basic objectives: **optimising cash flow movements** and **investing excess cash** — no single strategy achieves both.

Ways of optimising cash flows: accelerating inflows; managing blocked funds; leading and lagging; transfer pricing; netting; investing excess cash.

S.1.1 Accelerating Cash Inflows

- → Customers send cheques to **lockboxes** at various locations → less time and collection cost.
- → **Pre-authorised payment** — firm charges customer's bank account up to a limit.

S.1.2 Managing Blocked Funds

- → Host country may block remittance or require **local reinvestment** (jobs).
- → Subsidiary may **borrow locally for the parent** so blocked funds repay those bank loans.
- → Assess future blockage (political risk); move funds via **transfer pricing, direct negotiations, leading and lagging**.

S.1.3 Leading and Lagging

- → Adjust **timing of payments** (credit terms between units) by expected currency moves — reduces exposure or increases working capital.
- → **Lead** (accelerate) **hard currency payables**; **lag** (delay) **soft currency payables**.
- → SM example: Indian subsidiary buys from Japanese subsidiary. Rupee expected to **fall** vs yen → pay early (**leading**). Rupee expected to **strengthen** → delay (**lagging**).
- → Check government restrictions first.

S.1.4 Minimising Tax on Cash Flows through Transfer Pricing

- → Higher transfer price → **larger profit for transferor** division vs transferee.
- → For MNCs complicated by exchange restrictions, inflation differentials, import duties, **tax rate differentials**, host-country quotas. (Applied in TYK P2: $10 mn transfer price.)

5.1.5 Netting

- Optimises cash flows by combined effort of subsidiaries → lower **administrative and conversion costs**; minimises total inter-company fund flow.
- **Advantages:** fewer **cross-border transactions**; less **forex conversion**; better **cash forecasting** (net transfers at period end); **accurate reporting** and settlement.
- **Bilateral:** parent–subsidiary or two subsidiaries. X buys $20 mn from Y, Y buys $30 mn from X → gross $50 mn; Y pays X only **$10 mn**.
- **Multilateral:** each affiliate nets all receipts against all payments, pays/receives the balance. Needs **centralised communication** and disciplined subsidiaries. P → Q → R → P, $50 mn each → transfers **completely eliminated**.

Method: matrix of receivables and payables in a common currency → each affiliate's net receipt / net payment.

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SM netting matrix (US parent; France, Germany, UK, Italy; $ '000): receipts France 200, Germany 180, UK 210, Italy 190; payments France 240, Germany 130, UK 160, Italy 250; gross **780**. Net: Germany +50, UK +50 receive; France −40, Italy −60 pay → net transfers **100**, a reduction of **87%**.

5.1.6 Investing Excess Cash

- **Euro currency market:** Euro dollar deposits yield more than US bank deposits; also foreign T-bills/CP.
- Centralised **pooling** of subsidiaries' surplus → larger deposits, higher returns; separate investment account lets one subsidiary fund another without bank conversion charges.
- Converting pooled funds into one currency costs money → **cost-benefit analysis**.
- Choose currencies with higher interest or expected appreciation; foreign deposit if effective yield beats domestic after exchange risk (forecast rates); **diversify** across currencies to avoid big loss on sudden depreciation.

5.2 International Inventory Management

- MNC holds stock above EOQ — **stock piling** (vertical set-up; political bottlenecks; importing currency expected to depreciate).
- Decision: **cumulative carrying cost vs expected price rise** from exchange rate change; if probability of supply interruption is very high, stockpile even at higher cost.
- Longer **lead time** + customs → **re-order point earlier**; depends on quantity imported vs locally available.

5.3 International Receivables Management

Inter-firm sales	Intra-firm sales
Currency: exporter wants **strong**, importer **weak** currency; exporter may accept weak currency if it has **debt in that currency**. **Terms:** with weak-currency invoicing, exporter gives short credit; liberal credit if it can borrow against bills receivable or competition is keen (finances importer's inventory, expands sales).	Focus on **global allocation of resources**; parts made in different countries → large receivables. Timing does not affect the firm; cash-surplus unit pays early, cash-short unit late — **leads and lags**.

Exam steps — foreign project NPV (home currency approach): 1 forward rates by PPP (inflation) or IRP (interest), watch direct vs indirect quote → 2 foreign CF (inflate if real; tax; depreciation; WC; terminal) → 3 apply remittance restriction / withholding tax / home tax less credit → 4 convert each year → 5 add home-currency CF → 6 discount at home rate → 7 accept if NPV > 0.

Exam steps — foreign currency approach: 1 $1 + RP = (1 + k_{home}) \div (1 + i_{home})$ → 2 $k_f = (1 + RP)(1 + i_f) - 1$ → 3 discount foreign CF → 4 foreign NPV × spot.

Exam steps — APV: 1 forward rates (IRP) if foreign flows → 2 base NPV: operating CF at unlevered k^* → 3 PV of interest tax shields (and subsidies) at before-tax cost of debt → 4 APV = base NPV + PV shields (− issue costs / + usable blocked funds).

Exam steps — GDR: 1 gross issue = net ÷ (1 − flotation) → 2 price per GDR = MP × shares × (1 − discount) ÷ rate → 3 number = gross ÷ price → 4 $K_e = D_1$ per GDR ÷ net proceeds per GDR + g.

Exam steps — netting: 1 convert all to common currency → 2 matrix: receipts (row) vs payments (column) → 3 net position per affiliate → 4 net transfers vs gross, % saving.

11 International Financial Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

25 marks set across the last three attempts (7.1% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.

Attempt	Question	Marks	Concept tested
Sep 2025	Q5(a)	7	Foreign (Nepal) import project: NPV with NC cash flows converted at forward rates from the inflation differential and discounted at the Indian rate (Numerical)
Jan 2026	Q5(b)	6	GDR issue: number of GDRs for a dollar requirement and cost of GDR to the company (discount, flotation cost, dividend growth) (Numerical)
May 2026	Q1(a)	6	Foreign project with a currency swap at spot for the initial and terminal principal; forward rates for technical fees and the excess resale proceeds; PV of facilitation fee; NPV in rupees and accept/reject (Numerical)
May 2026	Q3(a)	6	Project in Europe with 50% debt: forward rates by interest rate parity, cheapest borrowing option, then APV (unlevered NPV at 12% plus PV of tax shield at 5%); ICAI gives a main and an alternative answer (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Write a short note on Instruments of International Finance.

ICAI answer: "Please refer paragraph 2" – SM paragraph 2 reproduced

2. INTERNATIONAL SOURCES OF FINANCE

Indian companies have been able to tap global markets to raise foreign currency funds by issuing various types of financial instruments which are discussed as follows:

2.1 Foreign Currency Convertible Bonds (FCCBs)

A type of convertible bond issued in a currency different than the issuer's domestic currency. In other words, the money being raised by the issuing company is in the form of a foreign currency. A convertible bond is a mix between a debt and equity instrument. It acts like a bond by making regular coupon and principal payments, but these bonds also give the bondholder the option to convert the bond into stock.

These types of bonds are attractive to both investors and issuers. The investors receive the safety of guaranteed payments on the bond and are also able to take advantage of any large price appreciation in the company's stock. (Bondholders take advantage of this appreciation by means of warrants attached to the bonds, which are activated when the price of the stock reaches a certain point.) Due to the equity side of the bond, which adds value, the coupon payments on the bond are lower for the company, thereby reducing its debt-financing costs.

Advantages of FCCBs

(i) The convertible bond gives the investor the flexibility to convert the bond into equity at a price or redeem the bond at the end of a specified period, normally three years if the price of the share has not met his expectations.

(ii) Companies prefer bonds as it leads to delayed dilution of equity and allows company to avoid any current dilution in earnings per share that a further issuance of equity would cause.

(iii) FCCBs are easily marketable as investors enjoy the option of conversion into equity resulting into capital appreciation. Further investor is assured of a minimum fixed interest earnings.

Disadvantages of FCCBs

(i) Exchange risk is more in FCCBs as interest on bonds would be payable in foreign currency. Thus, companies with low debt equity ratios, large forex earnings potential only opt for FCCBs.

(ii) FCCBs mean creation of more debt and a forex outgo in terms of interest which is in foreign exchange.

(iii) In the case of convertible bonds, the interest rate is low, say around 3–4% but there is exchange risk on the interest payment as well as re-payment if the bonds are not converted into equity shares. The only major advantage would be that where the company has a high

rate of growth in earnings and the conversion takes place subsequently, the price at which shares can be issued can be higher than the current market price.

2.2 American Depository Receipts (ADRs)

Depository receipts issued by a company in the United States of America (USA) is known as American Depository Receipts (ADRs). Such receipts must be issued in accordance with the provisions stipulated by the Securities and Exchange Commission of USA (SEC) which are very stringent.

An ADR is generally created by the deposit of the securities of a non-United States company with a custodian bank in the country of incorporation of the issuing company. The custodian bank informs the depository in the United States that the ADRs can be issued. ADRs are United States dollar denominated and are traded in the same way as are the securities of United States companies. The ADR holder is entitled to the same rights and advantages as owners of the underlying securities in the home country. Several variations on ADRs have developed over time to meet more specialized demands in different markets. One such variation is the GDR which are identical in structure to an ADR, the only difference being that they can be traded in more than one currency and within as well as outside the United States.

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Through the issue of depository receipts, companies in India have been able to tap global equity market to raise foreign currency funds by way of equity. Quite apart from the specific needs that Indian companies may have for equity capital in preference to debt and the perceived advantages of raising equity over debt in general (no repayment of "principal" and generally lower servicing costs, etc.) the fact of the matter is quite simple, that no other form of term foreign exchange funding has been available. In addition, it has been perceived that a GDR issue has been able to fetch higher prices from international investors (even when Indian issues were being sold at a discount to the prevailing domestic share prices) than those that a domestic public issue would have been able to extract from Indian investors.

- **Impact of GDRs on Indian Capital Market**

  Since the inception of GDRs a remarkable change in Indian capital market has been observed as follows:

  (i) Indian stock market to some extent is shifting from Bombay to Luxemburg.

  (ii) There is arbitrage possibility in GDR issues.

  (iii) Indian stock market is no longer independent from the rest of the world. This puts additional strain on the investors as they now need to keep updated with world wide economic events.

  (iv) Indian retail investors are completely sidelined. GDRs/Foreign Institutional Investors' placements + free pricing implies that retail investors can no longer expect to make easy money on heavily discounted rights/public issues.

  As a result of introduction of GDRs a considerable foreign investment has flown into India.

- **Markets of GDRs**

  (i) GDR's are sold primarily to institutional investors.

  (ii) Demand is likely to be dominated by emerging market funds.

  (iii) Switching by foreign institutional investors from ordinary shares into GDRs is likely.

  (iv) Major demand is also in UK, USA (Qualified Institutional Buyers), South East Asia (Hong Kong, Singapore), and to some extent continental Europe (principally France and Switzerland).

- **Mechanism of GDR:** The mechanics of a GDR issue may be described with the help of following diagram.



Characteristics

(i) Holders of GDRs participate in the economic benefits of being ordinary shareholders, though they do not have voting rights.

(ii) GDRs are settled through CEDEL & Euro-clear international book entry systems.

(iii) GDRs are listed on the Luxemburg stock exchange.

(iv) Trading takes place between professional market makers on an OTC (over the counter) basis.

(v) The instruments are freely traded.

(vi) They are marketed globally without being confined to borders of any market or country as it can be traded in more than one currency.

(vii) Investors earn fixed income by way of dividends which are paid in issuer currency converted into dollars by depository and paid to investors and hence exchange risk is with investor.

(viii) As far as the case of liquidation of GDRs is concerned, an investor may get the GDR cancelled any time after a cooling period of 45 days. A non-resident holder of GDRs may ask the overseas bank (depository) to redeem (cancel) the GDRs. In that case overseas depository bank shall request the domestic custodians bank to cancel the GDR and to get the corresponding underlying shares released in favour of non-resident investor. The price of the ordinary shares of the issuing company prevailing in the Bombay Stock Exchange or the National Stock Exchange on the date of advice of redemption shall be taken as the cost of acquisition of the underlying ordinary share.

Illustration 4

X Ltd. is interested in expanding its operation and planning to install manufacturing plant at US. For the proposed project it requires a fund of $ 10 million (net of issue expenses/ floatation cost). The estimated floatation cost is 2%. To finance this project it proposes to issue GDRs.

You as financial consultant is required to compute the number of GDRs to be issued and cost of the GDR with the help of following additional information.

(i) Expected market price of share at the time of issue of GDR is ₹ 250 (Face Value ₹ 100)

(ii) 2 Shares shall underly each GDR and shall be priced at 10% discount to market price.

(iii) Expected exchange rate ₹ 60/$.

(iv) Dividend expected to be paid is 20% with growth rate 12%.

Solution

Net Issue Size = $10 million

Gross Issue = $\frac{\$10\,\text{million}}{0.98}$ = $ 10.204 million

Issue Price per GDR in ₹ (250 x 2 x 90%)	₹ 450
Issue Price per GDR in $ (₹ 450/ ₹ 60)	$ 7.50
Dividend Per GDR (D_1) (₹ 20 x 2)	₹ 40
Net Proceeds Per GDR (₹ 450 x 0.98)	₹ 441.00

(a) Number of GDR to be issued $\frac{\$10.204 \text{ million}}{\$7.50}$ = 1.3605 million

(b) Cost of GDR to X Ltd. $k_e = \frac{40.00}{441.00} + 0.12 = 21.07\%$

2.4 Euro-Convertible Bonds (ECBs)

A convertible bond is a debt instrument which gives the holders of the bond an option to convert the bond into a predetermined number of equity shares of the company. Usually, the price of the equity shares at the time of conversion will have a premium element. The bonds carry a fixed rate of interest. If the issuer company desires, the issue of such bonds may carry two options viz.

(i) Call Options: (Issuer's option) - where the terms of issue of the bonds contain a provision for call option, the issuer company has the option of calling (buying) the bonds for redemption before the date of maturity of the bonds. Where the issuer's share price has appreciated substantially, i.e. far in excess of the redemption value of the bonds, the issuer company can exercise the option. This call option forces the investors to convert the bonds into equity. Usually, such a case arises when the share prices reach a stage near 130% to 150% of the conversion price.

(ii) Put options: (Holder's option) - A provision of put option gives the holder of the bonds a right to put (sell)his bonds back to the issuer company at a pre-determined price and date. In case of Euro-convertible bonds, the payment of interest on and the redemption of the bonds will be made by the issuer company in US dollars.

2.5 Other Sources

- ***Euro-Convertible Zero Coupon Bonds:*** These bonds are structured as a convertible bond. No interest is payable on the bonds. But conversion of bonds takes place on maturity at a pre-determined price. Usually there is a 5 years maturity period and they are treated as a deferred equity issue.
- ***Euro-bonds with Equity Warrants*:** These bonds carry a coupon rate determined by the market rates. The warrants are detachable. Pure bonds are traded at a discount. Fixed income funds may like to invest in these bonds for the purposes of regular income.
- ***Syndicated bank loans:*** It is one of the earlier ways of raising funds in the form of large loans from banks with good credit rating, can be arranged in reasonably short time and with few formalities. The maturity of the loan can be for a duration of 5 to 10 years. The interest rate is generally set with reference to an index, say, SOFR plus a spread which depends upon the credit rating of the borrower. Some covenants are laid down by the lending institution like maintenance of key financial ratios.
- ***Euro-bonds*:** These are basically debt instruments denominated in a currency issued outside the country of that currency for examples Yen bond floated in France. Primary attraction of these bonds is the refuge from tax and regulations and provide scope for arbitraging yields. These are usually bearer bonds and can take the form of

  (i) Traditional Fixed Rate Bonds.

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- ***Foreign Bonds*:** Foreign bonds are denominated in a currency which is foreign to the borrower and sold at the country of that currency. Such bonds are always subject to the restrictions and are placed by that country on the foreigners funds.
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- ***Credit Instruments:*** The foregoing discussion relating to foreign exchange risk management and international capital market shows that foreign exchange operations of banks consist primarily of purchase and sale of credit instruments. There are many types of credit instruments used in effecting foreign remittances. They differ in the speed, with which money can be received by the creditor at the other end after it has been paid in by the debtor at his end. The price or the rate of each instrument, therefore, varies with extent of the loss of interest and risk of loss involved. There are, therefore, different rates of exchange applicable to different types of credit instruments.

ICAI answer: "Please refer paragraph 2" – SM paragraph 2 reproduced

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Theory Q2

2. What is the impact of GDRs on Indian Capital Market?

ICAI answer: "Please refer paragraph 2.3" – SM paragraph 2.3 reproduced

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(v) The instruments are freely traded.

(vi) They are marketed globally without being confined to borders of any market or country as it can be traded in more than one currency.

(vii) Investors earn fixed income by way of dividends which are paid in issuer currency converted into dollars by depository and paid to investors and hence exchange risk is with investor.

(viii) As far as the case of liquidation of GDRs is concerned, an investor may get the GDR cancelled any time after a cooling period of 45 days. A non-resident holder of GDRs may ask the overseas bank (depository) to redeem (cancel) the GDRs. In that case overseas depository bank shall request the domestic custodians bank to cancel the GDR and to get the corresponding underlying shares released in favour of non-resident investor. The price of the ordinary shares of the issuing company prevailing in the Bombay Stock Exchange or the National Stock Exchange on the date of advice of redemption shall be taken as the cost of acquisition of the underlying ordinary share.

Illustration 4

X Ltd. is interested in expanding its operation and planning to install manufacturing plant at US. For the proposed project it requires a fund of $ 10 million (net of issue expenses/ floatation cost). The estimated floatation cost is 2%. To finance this project it proposes to issue GDRs.

You as financial consultant is required to compute the number of GDRs to be issued and cost of the GDR with the help of following additional information.

(i) Expected market price of share at the time of issue of GDR is ₹ 250 (Face Value ₹ 100)

(ii) 2 Shares shall underly each GDR and shall be priced at 10% discount to market price.

(iii) Expected exchange rate ₹ 60/$.

(iv) Dividend expected to be paid is 20% with growth rate 12%.

Solution

Net Issue Size = $10 million

Gross Issue = $\frac{\$10\,\text{million}}{0.98}$ = $ 10.204 million

Issue Price per GDR in ₹ (250 x 2 x 90%)	₹ 450
Issue Price per GDR in $ (₹ 450/ ₹ 60)	$ 7.50
Dividend Per GDR (D_1) (₹ 20 x 2)	₹ 40
Net Proceeds Per GDR (₹ 450 x 0.98)	₹ 441.00

(a) Number of GDR to be issued $\frac{\$10.204\ \text{million}}{\$7.50}$ = 1.3605 million

(b) Cost of GDR to X Ltd. $k_e = \frac{40.00}{441.00} + 0.12 = 21.07\%$

ICAI answer: "Please refer paragraph 2.3" – SM paragraph 2.3 reproduced

2.3 Global Depository Receipts (GDRs)

A depository receipt is basically a negotiable certificate, denominated in a currency not native to the issuer, that represents the company's publicly - traded local currency equity shares. Most GDRs are denominated in USD, while a few are denominated in Euro and Pound Sterling. The Depository Receipts issued in the US are called American Depository Receipts (ADRs), which anyway are denominated in USD and outside of USA, these are called GDRs In theory, though a depository receipt can also represent a debt instrument, in practice it rarely does. DRs (depository receipts) are created when the local currency shares of an Indian company are delivered to the depository's local custodian bank, against which the Depository bank (such as the Bank of New York) issues depository receipts in US dollar. These depository receipts may trade freely in the overseas markets like any other dollar-denominated security, either on a foreign stock exchange, or in the over-the-counter market, or among a restricted group such as Qualified Institutional Buyers (QIBs). Indian issues have taken the form of GDRs to reflect the fact that they are marketed globally, rather than in a specific country or market.

Through the issue of depository receipts, companies in India have been able to tap global equity market to raise foreign currency funds by way of equity. Quite apart from the specific needs that Indian companies may have for equity capital in preference to debt and the perceived advantages of raising equity over debt in general (no repayment of "principal" and generally lower servicing costs, etc.) the fact of the matter is quite simple, that no other form of term foreign exchange funding has been available. In addition, it has been perceived that a GDR issue has been able to fetch higher prices from international investors (even when Indian issues were being sold at a discount to the prevailing domestic share prices) than those that a domestic public issue would have been able to extract from Indian investors.

- **Impact of GDRs on Indian Capital Market**

  Since the inception of GDRs a remarkable change in Indian capital market has been observed as follows:

  (i) Indian stock market to some extent is shifting from Bombay to Luxemburg.

  (ii) There is arbitrage possibility in GDR issues.

  (iii) Indian stock market is no longer independent from the rest of the world. This puts additional strain on the investors as they now need to keep updated with world wide economic events.

  (iv) Indian retail investors are completely sidelined. GDRs/Foreign Institutional Investors' placements + free pricing implies that retail investors can no longer expect to make easy money on heavily discounted rights/public issues.

  As a result of introduction of GDRs a considerable foreign investment has flown into India.

- **Markets of GDRs**

  (i) GDR's are sold primarily to institutional investors.

  (ii) Demand is likely to be dominated by emerging market funds.

  (iii) Switching by foreign institutional investors from ordinary shares into GDRs is likely.

  (iv) Major demand is also in UK, USA (Qualified Institutional Buyers), South East Asia (Hong Kong, Singapore), and to some extent continental Europe (principally France and Switzerland).

- **Mechanism of GDR:** The mechanics of a GDR issue may be described with the help of following diagram.



Characteristics

(i) Holders of GDRs participate in the economic benefits of being ordinary shareholders, though they do not have voting rights.

(ii) GDRs are settled through CEDEL & Euro-clear international book entry systems.

(iii) GDRs are listed on the Luxemburg stock exchange.

(iv) Trading takes place between professional market makers on an OTC (over the counter) basis.

(v) The instruments are freely traded.

(vi) They are marketed globally without being confined to borders of any market or country as it can be traded in more than one currency.

(vii) Investors earn fixed income by way of dividends which are paid in issuer currency converted into dollars by depository and paid to investors and hence exchange risk is with investor.

(viii) As far as the case of liquidation of GDRs is concerned, an investor may get the GDR cancelled any time after a cooling period of 45 days. A non-resident holder of GDRs may ask the overseas bank (depository) to redeem (cancel) the GDRs. In that case overseas depository bank shall request the domestic custodians bank to cancel the GDR and to get the corresponding underlying shares released in favour of non-resident investor. The price of the ordinary shares of the issuing company prevailing in the Bombay Stock Exchange or the National Stock Exchange on the date of advice of redemption shall be taken as the cost of acquisition of the underlying ordinary share.

Illustration 4

X Ltd. is interested in expanding its operation and planning to install manufacturing plant at US. For the proposed project it requires a fund of $ 10 million (net of issue expenses/ floatation cost). The estimated floatation cost is 2%. To finance this project it proposes to issue GDRs.

You as financial consultant is required to compute the number of GDRs to be issued and cost of the GDR with the help of following additional information.

(i) Expected market price of share at the time of issue of GDR is ₹ 250 (Face Value ₹ 100)

(ii) 2 Shares shall underly each GDR and shall be priced at 10% discount to market price.

(iii) Expected exchange rate ₹ 60/$.

(iv) Dividend expected to be paid is 20% with growth rate 12%.

Solution

Net Issue Size = $10 million

Gross Issue = $\frac{\$10\,\text{million}}{0.98}$ = $ 10.204 million

Issue Price per GDR in ₹ (250 x 2 x 90%)	₹ 450
Issue Price per GDR in $ (₹ 450/ ₹ 60)	$ 7.50
Dividend Per GDR (D_1) (₹ 20 x 2)	₹ 40
Net Proceeds Per GDR (₹ 450 x 0.98)	₹ 441.00

(a) Number of GDR to be issued $\frac{\$10.204 \text{ million}}{\$7.50}$ = 1.3605 million

(b) Cost of GDR to X Ltd. $k_e = \frac{40.00}{441.00} + 0.12 = 21.07\%$

① SM illustrations with ICAI solutions

I. International Capital Budgeting

SM Ch 11 · Illustration 1 (under 1.6.1 A foreign company is investing in India)

Illustration 1

Perfect Inc., a U.S. based Pharmaceutical Company has received an offer from Aidscure Ltd., a company engaged in manufacturing of drugs to cure Dengue, to set up a manufacturing unit in Baddi (H.P.), India in a joint venture.

As per the Joint Venture agreement, Perfect Inc. will receive 55% share of revenues plus a royalty @ US $0.01 per bottle. The initial investment will be ₹ 200 crores for machinery and factory. The scrap value of machinery and factory is estimated at the end of five (5) year to be ₹ 5 crores. The machinery is depreciable @ 20% on the value net of salvage value using Straight Line Method. An initial working capital to the tune of ₹ 50 crores shall be required and thereafter ₹ 5 crores each year.

As per GOI directions, it is estimated that the price per bottle will be ₹ 7.50 and production will be 24 crores bottles per year. The price in addition to inflation of respective years shall be increased by ₹ 1 each year. The production cost shall be 40% of the revenues.

The applicable tax rate in India is 30% and 35% in US and there is Double Taxation Avoidance Agreement between India and US. According to the agreement tax credit shall be given in US for the tax paid in India. In both the countries, taxes shall be paid in the following year in which profit have arisen/ remittance received.

The Spot rate of $ is ₹ 57. The inflation in India is 6% (expected to decrease by 0.50% every year) and 5% in US.

As per the policy of GOI, only 50% of the share can be remitted in the year in which they are realised and remaining in the following year.

Though WACC of Perfect Inc. is 13% but due to risky nature of the project it expects a return of 15%.

Determine whether Perfect Inc. should invest in the project or not (from subsidiary point of view).

Solution — ICAI

Solution

Working Notes:

1. Estimated Exchange Rates (Using PPP Theory)

Year	0	1	2	3	4	5	6
Exchange rate *	57	57.54	57.82	57.82	57.54	56.99	56.18

2. Share in sales

Year	1	2	3	4	5
Annual Units in crores	24	24	24	24	24
Price per bottle (₹)	7.50	8.50	9.50	10.50	11.50
Price fluctuating Inflation Rate	6.00%	5.50%	5.00%	4.50%	4.00%
Inflated Price (₹)	7.95	8.97	9.98	10.97	11.96
Inflated Sales Revenue (₹ Crore)	190.80	215.28	239.52	263.28	287.04
Sales share @55%	104.94	118.40	131.74	144.80	157.87

3. Royalty Payment

Year	1	2	3	4	5
Annual Units in crores	24	24	24	24	24
Royalty in $	0.01	0.01	0.01	0.01	0.01
Total Royalty ($ Crore)	0.24	0.24	0.24	0.24	0.24
Exchange Rate	57.54	57.82	57.82	57.54	56.99
Total Royalty (₹ Crore)	13.81	13.88	13.88	13.81	13.68

4. Tax Liability

(₹ Crore)

Year	1	2	3	4	5
Sales Share	104.94	118.40	131.74	144.80	157.87
Total Royalty	13.81	13.88	13.88	13.81	13.68
Total Income	118.75	132.28	145.61	158.61	171.55
Less: Expenses					
Production Cost (Sales share x 40%)	41.98	47.36	52.69	57.92	63.15
Depreciation (195 x 20%)	39.00	39.00	39.00	39.00	39.00
PBT	37.77	45.92	53.92	61.69	69.40
Tax on Profit @30%	11.33	13.78	16.18	18.51	20.82
Net Profit	26.44	32.14	37.74	43.18	48.58

5. Cash Flow

(₹ Crore)

Year	0	1	2	3	4	5	6
Sales Share	0.00	104.94	118.40	131.74	144.80	157.87	0.00
Total Royalty	0.00	13.81	13.88	13.88	13.81	13.68	0.00
Production Cost	0.00	-41.98	-47.36	-52.69	-57.92	-63.15	0.00
Initial Outlay	-200.00	0.00	0.00	0.00	0.00	0.00	0.00
Working Capital	-50.00	-5.00	-5.00	-5.00	-5.00	70.00	0.00
Scrap Value	0.00	0.00	0.00	0.00	0.00	5.00	0.00
Tax on Profit	0.00	0.00	-11.33	-13.78	-16.18	-18.51	-20.82
Free Cash Flow	**-250.00**	**71.77**	**68.59**	**74.15**	**79.51**	**164.89**	**-20.82**

6. Remittance of Cash Flows (₹ Crore)

Year	0	1	2	3	4	5	6
Free Cash Flow	-250.00	71.77	68.59	74.15	79.51	164.89	-20.82
50% of Current Year Cash Flow	0.00	35.89	34.29	37.07	39.76	82.45	0.00
Previous year remaining cash flow	0.00	0.00	35.88	34.30	37.08	39.75	82.44
Total Remittance	**-250.00**	**35.88**	**70.17**	**71.37**	**76.84**	**122.20**	**61.62**

NPV of Project under Appraisal

Year	0	1	2	3	4	5	6	7
Total Remittance (₹ Crore)	-250.00	35.88	70.17	71.37	76.84	122.20	61.62	-
Exchange Rate	57.00	57.54	57.82	57.82	57.54	56.99	56.18	-
Remittance ($ mn)	-43.86	6.24	12.14	12.34	13.35	21.44	10.97	-
US Tax @35% ($ mn)	0.00	0.00	2.18	4.25	4.32	4.67	7.50	3.84
Indian Tax ($ mn)	0.00	0.00	1.96	2.38	2.82	3.25	3.71	-
Net Tax ($ mn)	0.00	0.00	0.22	1.87	1.51	1.42	3.79	3.84
Net Cash Flow ($ mn)	-43.86	6.24	11.92	10.47	11.84	20.02	7.18	-3.84
PVF @ 15%	1.000	0.870	0.756	0.658	0.572	0.497	0.432	0.376
Present Value ($ mn)	-43.86	5.43	9.01	6.89	6.77	9.95	3.10	-1.44
Net Present Value ($ mn)								= -4.15

Decision: Since NPV of the project is negative, Perfect inc. should not invest in the project.

* Estimated exchange rates have been calculated by using the following formula:

Expected spot rate = Current Spot Rate x expected difference in inflation rates

$$E(S_1) = S_0 \times \frac{(1 + I_d)}{(1 + 1_f)}$$

Where

$E(S_1)$ is the expected Spot rate in time period 1

S_0 is the current spot rate (Direct Quote)

I_d is the inflation in the domestic country (home country)

I_f is the inflation in the foreign country

SM Ch 11 · Illustration 2 (under 1.6.2 An Indian Company is investing in foreign country by r)

Illustration 2

Its Entertainment Ltd., an Indian Amusement Company is happy with the success of its Water Park in India. The company wants to repeat its success in Nepal also where it is planning to establish a Grand Water Park with world class amenities. The company is also encouraged by a marketing research report on which it has just spent ₹ 20,00,000.

The estimated cost of construction would be Nepali Rupee (NPR) 450 crores and it would be completed in one years time. Half of the construction cost will be paid in the beginning and rest at the end of year. In addition, working capital requirement would be NPR 65 crores from the year end one. The after tax realizable value of fixed assets after four years of operation is expected to be NPR 250 crores. Under the Foreign Capital Encouragement Policy of Nepal, company is allowed to claim 20% depreciation allowance per year on reducing balance basis subject to maximum capital limit of NPR 200 crore. The company can raise loan for theme park in Nepal @ 9%.

The water park will have a maximum capacity of 20,000 visitors per day. On an average, it is expected to achieve 70% capacity for first operational four years. The entry ticket is expected to be NPR 220 per person. In addition to entry tickets revenue, the company could earn revenue from sale of food and beverages and fancy gift items. The average sales expected to be NPR 150 per visitor for food and beverages and NPR 50 per visitor for fancy gift items. The sales margin on food and beverages and fancy gift items is 20% and 50% respectively. The park would open for 360 days a year.

The annual staffing cost would be NPR 65 crores per annum. The annual insurance cost would be NPR 5 crores. The other running and maintenance costs are expected to be NPR 25 crores in the first year of operation which is expected to increase NPR 4 crores every year. The company would apportion existing overheads to the tune of NPR 5 crores to the park.

All costs and receipts (excluding construction costs, assets realizable value and other running and maintenance costs) mentioned above are at current prices (i.e. 0 point of time) which are expected to increase by 5% per year.

The current spot rate is NPR 1.60 per rupee. The tax rate in India is 30% and in Nepal it is 20%.

The average market return is 11% and interest rate on treasury bond is 8%. The company's current equity beta is 0.45. The company's funding ratio for the Water Park would be 55% equity and 45% debt.

Being a tourist Place, the amusement industry in Nepal is competitive and very different from its Indian counterpart. The company has gathered the relevant information about its nearest competitor in Nepal. The competitor's market value of the equity is NPR 1850 crores and the debt is NPR 510 crores and the equity beta is 1.35.

State whether Its Entertainment Ltd. should undertake Water Park project in Nepal or not.

Solution — ICAI

Solution

Working Notes:

1. Calculation of Cost of Funds/ Discount Rate

Competing Company's Information	
Equity Market Value	1850.00
Debt Market Value	510.00
Equity Beta	1.35

Assuming debt to be risk free i.e. beta is zero, the beta of competitor is un-geared as follows:

$$\text{Asset Beta} = \text{Equity Beta} \times \frac{E}{E + D(1-t)} = 1.35 \times \frac{1850}{1850 + 510(1-0.20)} = 1.106$$

Equity beta for Its Entertainment Ltd. in Nepal

Assets beta in Nepal	1.106
Ratio of funding in Nepal	
Equity	55.00%
Debt	45.00%

1. $1.106 = \text{Equity Beta} \times \frac{55}{55 + 45(1 - 0.20)}$

Equity Beta = 1.83

Cost of Equity as per CAPM

Market Return 11.00%

Risk free return 8.00%

Cost of Equity = Risk free return + β (Market Return - Risk free return)

= 8.00% + 1.83(11.00% - 8.00%) = 13.49%

WACC = 13.49% x 0.55 + 9%(1- 0.20) x 0.45 = 10.66%

2. Present Value Factors at the discount rate of 10.66%

Year	0	1	2	3	4	5
PVAF	1.000	0.904	0.817	0.738	0.667	0.603

3. Calculation of Capital Allowances

Year	1	2	3	4
Opening Balance (NPR Crore)	200.00	160.00	128.00	102.40
Less: Depreciation (NPR Crore)	40.00	32.00	25.60	20.48
Closing Balance (NPR Crore)	160.00	128.00	102.40	81.92

Calculation of Present of Free Cash Flow

Year	0	1	2	3	4	5
Expected Annual visitors			5040000	5040000	5040000	5040000
Entry ticket price per visitor (NPR)			242.55	254.68	267.41	280.78
Profit from sale of Food and Beverages per visitor (NPR)			33.08	34.73	36.47	38.29
Profit from sale of Fancy Gift Items per visitor (NPR)			27.56	28.94	30.39	31.91
Revenue per visitor (NPR)			303.19	318.35	334.27	350.98
Total Revenue (NPR crores)			152.81	160.45	168.47	176.89
Less:						
Annual Staffing Cost (NPR crores)			71.66	75.25	79.01	82.96
Annual Insurance Costs (NPR crores)			5.51	5.79	6.08	6.38
Other running and maintenance costs (NPR crores)			25.00	29.00	33.00	37.00
Depreciation Allowances (NPR crores)			40.00	32.00	25.60	20.48

Total Expenses (NPR crores)			142.17	142.04	143.69	146.82
PBT (NPR crores)			10.64	18.41	24.78	30.07
Tax on Profit (NPR crores)			2.13	3.68	4.96	6.01
Net Profit (NPR crores)			8.51	14.73	19.82	24.06
Add: Depreciation Allowances (NPR crores)			40	32	25.6	20.48
Park Construction Cost (NPR crores)	-225	-225				
After tax assets realisation value (NPR crores)						250
Working capital (NPR crores)		-65.00	-3.25	-3.41	-3.58	75.25
Net cash Flow (NPR crores)	-225.00	-290.00	45.26	43.32	41.84	369.79
PVF at discount rate	1.00	0.904	0.817	0.738	0.667	0.603
Present Values (NPR crores)	-225.00	-262.16	36.98	31.97	27.91	222.98

Net Present Value (NPR crores) -167.32

Decision: Since NPV of the project is negative the Entertainment Ltd. should not undertake Water Park project in Nepal.

SM Ch 11 · Illustration 3 (under 1.6.3 An Indian Company is investing in foreign country by r)

Illustration 3

Opus Technologies Ltd., an Indian IT company is planning to make an investment through a wholly owned subsidiary in a software project in China with a shelf life of two years. The inflation in China is estimated as 8 percent. Operating cash flows are received at the year end.

For the project an initial investment of Chinese Yuan (CN¥) 30,00,000 will be in land. The land will be sold after the completion of project at estimated value of CN¥ 35,00,000. The project also requires an office complex at cost of CN¥ 15,00,000 payable at the beginning of project. The complex will be depreciated on straight-line basis over two years to a zero salvage value. This complex is expected to fetch CN¥ 5,00,000 at the end of project.

The company is planning to raise the required funds through GDR issue in Mauritius. Each GDR will have 5 common equity shares of the company as underlying security which are currently trading at ₹ 200 per share (Face Value = ₹10) in the domestic market. The company has currently paid the dividend of 25% which is expected to grow at 10% p.a. The total issue cost is estimated to be 1 percent of issue size.

The annual sales is expected to be 10,000 units at the rate of CN¥ 500 per unit. The price of unit is expected to rise at the rate of inflation. Variable operating costs are 40 percent of sales. Fixed operating costs will be CN¥ 22,00,000 per year and expected to rise at the rate of inflation.

The tax rate applicable in China for income and capital gain is 25 percent and as per GOI Policy no further tax shall be payable in India. The current spot rate of CN¥ 1 is ₹ 9.50. The nominal interest rate in India and China is 12% and 10% respectively and the international parity conditions hold

You are required to

(a) Identify expected future cash flows in China and determine NPV of the project in CN¥.

(b) Determine whether Opus Technologies should go for the project or not assuming that there neither there is restriction on the transfer of funds from China to India nor any charges/taxes payable on the transfer of funds.

Solution — ICAI

Solution

Working Notes:

1. Calculation of Cost of Capital (GDR)

Current Dividend (D_0)	2.50
Expected Dividend (D_1)	2.75
Net Proceeds (₹ 200 per share – 1%)	198.00
Growth Rate	10.00%

$$k_e = \frac{2.75}{198} + 0.10 = 0.1139 \text{ i.e. } 11.39\%$$

2. Calculation of Expected Exchange Rate as per Interest Rate Parity

YEAR	EXPECTED RATE
1	$= 9.50 \times \frac{(1+0.12)}{(1+0.10)} = 9.67$

2	$= 9.50 \times \frac{(1+0.12)^2}{(1+0.10)^2} = 9.85$

3. Realization on the disposal of Land net of Tax

	CN¥
Sale value at the end of project	3500000.00
Cost of Land	3000000.00
Capital Gain	500000.00
Tax paid	125000.00
Amount realized net of tax	3375000.00

4. Realization on the disposal of Office Complex

	(CN¥)
Sale value at the end of project	500000.00
WDV	0.00
Capital Gain	500000.00
Tax paid	125000.00
Amount realized net of tax (A)	375000.00

5. Computation of Annual Cash Inflows

Year	1	2
Annual Units	10000	10000
Price per bottle (CN¥)	540.00	583.20
Annual Revenue (CN¥)	5400000.00	5832000.00
Less: Expenses		
Variable operating cost (CN¥)	2160000.00	2332800.00
Depreciation (CN¥)	750000.00	750000.00
Fixed Cost per annum (CN¥)	2376000.00	2566080.00
PBT (CN¥)	114000.00	183120.00
Tax on Profit (CN¥)	28500.00	45780.00
Net Profit (CN¥)	85500.00	137340.00
Add: Depreciation (CN¥)	750000.00	750000.00
Cash Flow	835500.00	887340.00

(a) Computation of NPV of the project in CN¥

(CN¥)

Year	0	1	2
Initial Investment	-4500000.00		
Annual Cash Inflows		835500.00	887340.00
Realization on the disposal of Land net of Tax			3375000.00
Realization on the disposal of Office Complex			375000.00
Total	-4500000.00	835500.00	4637340.00
PVF @11.39%	1.000	0.898	0.806
PV of Cash Flows	-4500000.00	750279.00	3737696.00
NPV			-12,025

(b) Evaluation of Project from Opus Point of View

(i) Assuming that inflow funds are transferred in the year in which same are generated i.e. first year and second year.

Year	0	1	2
Cash Flows (CN¥)	-4500000.00	835500.00	4637340.00
Exchange Rate (₹/ CN¥)	9.50	9.67	9.85
Cash Flows (₹)	-42750000.00	8079285.00	45677799.00
PVF @ 12%	1.00	0.893	0.797
	-42750000.00	7214802.00	36405206.00
NPV			870008.00

(ii) Assuming that inflow funds are transferred at the end of the project i.e. second year.

Year	0	2
Cash Flows (CN¥)	-4500000.00	5472840.00
Exchange Rate (₹/ CN¥)	9.50	9.85
Cash Flows (₹)	-42750000.00	53907474.00
PVF	1.00	0.797
	-42750000.00	42964257.00
NPV		214257.00

Though in terms of CN¥ the NPV of the project is negative but in ₹ it has positive NPV due to weakening of ₹ in comparison of CN¥. Thus, Opus can accept the project.

2. International Sources Of Finance

SM Ch 11 · Illustration 4 (under 2.3 Global Depository Receipts (GDRs))

Illustration 4

X Ltd. is interested in expanding its operation and planning to install manufacturing plant at US. For the proposed project it requires a fund of $ 10 million (net of issue expenses/ floatation cost). The estimated floatation cost is 2%. To finance this project it proposes to issue GDRs.

You as financial consultant is required to compute the number of GDRs to be issued and cost of the GDR with the help of following additional information.

(i) Expected market price of share at the time of issue of GDR is ₹ 250 (Face Value ₹ 100)

(ii) 2 Shares shall underly each GDR and shall be priced at 10% discount to market price.

(iii) Expected exchange rate ₹ 60/$.

(iv) Dividend expected to be paid is 20% with growth rate 12%.

Solution — ICAI

Solution

Net Issue Size = $10 million

Gross Issue = $\frac{\$10\,\text{million}}{0.98}$ = $ 10.204 million

Issue Price per GDR in ₹ (250 x 2 x 90%)	₹ 450
Issue Price per GDR in $ (₹ 450/ ₹ 60)	$ 7.50
Dividend Per GDR (D_1) (₹ 20 x 2)	₹ 40
Net Proceeds Per GDR (₹ 450 x 0.98)	₹ 441.00

(a) Number of GDR to be issued $\frac{\$10.204\ \text{million}}{\$7.50}$ = 1.3605 million

(b) Cost of GDR to X Ltd. $k_e = \frac{40.00}{441.00} + 0.12 = 21.07\%$

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. ABC Ltd. is considering a project in US, which will involve an initial investment of US $ 1,10,00,000. The project will have 5 years of life. Current spot exchange rate is ₹ 48 per US $. The risk free rate in US is 8% and the same in India is 12%. Cash inflow from the project is as follows:

Year	Cash inflow
1	US $ 20,00,000
2	US $ 25,00,000
3	US $ 30,00,000
4	US $ 40,00,000
5	US $ 50,00,000

Calculate the NPV of the project using foreign currency approach. Required rate of return on this project is 14%.

Solution — ICAI

1. (1 + 0.12) (1 + Risk Premium) = (1 + 0.14)

Or, 1 + Risk Premium = 1.14/1.12 = 1.0179

Therefore, Risk adjusted dollar rate is = 1.0179 x 1.08 = 1.099 – 1 = 0.099

Calculation of NPV

Year	Cash flow (Million) US$	PV Factor at 9.9%	P.V.
1	2.00	0.910	1.820
2	2.50	0.828	2.070
3	3.00	0.753	2.259
4	4.00	0.686	2.744
5	5.00	0.624	3.120
			12.013
		Less: Investment	11.000
		NPV	1.013

Therefore, Rupee NPV of the project is = ₹ (48 x 1.013) Million= ₹48.624 Million

Practical Q2

2. A USA based company is planning to set up a software development unit in India. Software developed at the Indian unit will be bought back by the US parent at a transfer price of US $10 millions. The unit will remain in existence in India for one year; the software is expected to get developed within this time frame.

The US based company will be subject to corporate tax of 30 per cent and a withholding tax of 10 per cent in India and will not be eligible for tax credit in the US. The software developed will be sold in the US market for US $ 12.0 millions. Other estimates are as follows:

Rent for fully furnished unit with necessary hardware in India	₹ 15,00,000
Man power cost (80 software professional will be working for 10 hours each day)	₹ 400 per man hour
Administrative and other costs	₹ 12,00,000

Advise the US Company on the financial viability of the project. The rupee-dollar rate is ₹48/$.

Note: Assume 365 days a year.

Solution — ICAI

2. Proforma profit and loss account of the Indian software development unit

	₹	₹
Revenue		48,00,00,000
Less: Costs:		
Rent	15,00,000	
Manpower (₹400 x 80 x 10 x 365)	11,68,00,000	
Administrative and other costs	12,00,000	11,95,00,000

Earnings before tax		36,05,00,000
Less: Tax		10,81,50,000
Earnings after tax		25,23,50,000
Less: Withholding tax(TDS)		2,52,35,000
Repatriation amount (in rupees)		22,71,15,000
Repatriation amount (in dollars)		$ 4.7 million

Advise: The cost of development software in India for the US based company is $5.268 million. As the USA based Company is expected to sell the software in the US at $12.0 million, it is advised to develop the software in India.

Alternatively, if is assumed that since foreign subsidiary has paid taxes it will not pay withholding taxes then solution will be as under:

	₹	₹
Revenue		48,00,00,000
Less: Costs:		
Rent	15,00,000	
Manpower (₹400 x 80 x 10 x 365)	11,68,00,000	
Administrative and other costs	12,00,000	11,95,00,000
Earnings before tax		36,05,00,000
Less: Tax		10,81,50,000
Earnings after tax		25,23,50,000
Repatriation amount (in rupees)		25,23,50,000
Repatriation amount (in dollars)		$ 5,257,292

Advise: The cost of development software in India for the US based company is $4.743 million. As the USA based Company is expected to sell the software in the US at $12.0 million, it is advised to develop the software in India.

Alternatively, if it assumed that first the withholding tax @ 10% is being paid and then its credit is taken in the payment of corporate tax then solution will be as follows:

	₹	₹
Revenue		48,00,00,000
Less: Costs:		
Rent	15,00,000	
Manpower (₹400 x 80 x 10 x 365)	11,68,00,000	
Administrative and other costs	12,00,000	11,95,00,000
Earnings before tax		36,05,00,000
Less: Withholding Tax		3,60,50,000
Earnings after Withholding tax @ 10%		32,44,50,000
Less: Corporation Tax net of Withholding Tax		7,21,00,000
Repatriation amount (in rupees)		25,23,50,000
Repatriation amount (in dollars)		$ 5,257,292

Advise: The cost of development software in India for the US based company is $4.743 million. As the USA based Company is expected to sell the software in the US at $12.0 million, it is advised to develop the software in India.

Practical Q3

3. XY Limited is engaged in large retail business in India. It is contemplating for expansion into a country of Africa by acquiring a group of stores having the same line of operation as that of India.

The exchange rate for the currency of the proposed African country is extremely volatile. Rate of inflation is presently 40% a year. Inflation in India is currently 10% a year. Management of XY Limited expects these rates likely to continue for the foreseeable future.

Estimated projected cash flows, in real terms, in India as well as African country for the first three years of the project are as follows:

	Year – 0	Year – 1	Year – 2	Year - 3
Cash flows in Indian ₹ (000)	-50,000	-1,500	-2,000	-2,500
Cash flows in African Rands (000)	-2,00,000	+50,000	+70,000	+90,000

XY Ltd. assumes the year 3 nominal cash flows will continue to be earned each year indefinitely. It evaluates all investments using nominal cash flows and a nominal discounting rate. The present exchange rate is African Rand 6 to ₹ 1.

You are required to calculate the net present value of the proposed investment considering the following:

(i) African Rand cash flows are converted into rupees and discounted at a risk adjusted rate.

(ii) All cash flows for these projects will be discounted at a rate of 20% to reflect it's high risk.

(iii) Ignore taxation.

	Year - 1	Year - 2	Year - 3
PVIF @ 20%	0.833	0.694	0.579

Solution — ICAI

3. Calculation of NPV

Year	0	1	2	3
Inflation factor in India	1.00	1.10	1.21	1.331
Inflation factor in Africa	1.00	1.40	1.96	2.744
Exchange Rate (as per IRP)	6.00	7.6364	9.7190	12.3696
Cash Flows in ₹ '000				
Real	-50000	-1500	-2000	-2500
Nominal (1)	-50000	-1650	-2420	-3327.50
Cash Flows in African Rand '000				
Real	-200000	50000	70000	90000
Nominal	-200000	70000	137200	246960
In Indian ₹ '000 (2)	-33333	9167	14117	19965
Net Cash Flow in ₹ '000 (1)+(2)	-83333	7517	11697	16637
PVF@20%	1	0.833	0.694	0.579
PV	-83333	6262	8118	9633

NPV of 3 years = -59320 (₹ '000)

NPV of Terminal Value = $\frac{16637}{0.20} \times 0.579 = 48164$ (₹ '000)

Total NPV of the Project = -59320 (₹ '000) + 48164 (₹ '000) = -11156 (₹ '000)

Practical Q4

4. A multinational company is planning to set up a subsidiary company in India (where hitherto it was exporting) in view of growing demand for its product and competition from other MNCs. The initial project cost (consisting of Plant and Machinery including installation) is estimated to be US$ 500 million. The net working capital requirements are estimated at US$ 50 million. The company follows straight line method of depreciation. Presently, the company is exporting two million units every year at a unit price of US$ 80, its variable cost per unit being US$ 40.

The Chief Financial Officer has estimated the following operating cost and other data in respect of proposed project:

(i) Variable operating cost will be US $ 20 per unit of production;

(ii) Additional cash fixed cost will be US $ 30 million p.a. and project's share of allocated fixed cost will be US $ 3 million p.a. based on principle of ability to share;

(iii) Production capacity of the proposed project in India will be 5 million units;

(iv) Expected useful life of the proposed plant is five years with no salvage value;

(v) Existing working capital investment for production & sale of two million units through exports was US $ 15 million;

(vi) Export of the product in the coming year will decrease to 1.5 million units in case the company does not open subsidiary company in India, in view of the presence of competing MNCs that are in the process of setting up their subsidiaries in India;

(vii) Applicable Corporate Income Tax rate is 35%, and

(viii) Required rate of return for such project is 12%.

Assuming that there will be no variation in the exchange rate of two currencies and all profits will be repatriated, as there will be no withholding tax, estimate Net Present Value (NPV) of the proposed project in India.

Present Value Interest Factors (PVIF) @ 12% for five years are as below:

Year	1	2	3	4	5
PVIF	0.8929	0.7972	0.7118	0.6355	0.5674

Solution — ICAI

4. Financial Analysis whether to set up the manufacturing units in India or not may be carried using NPV technique as follows:

I. Incremental Cash Outflows

	$ Million
Cost of Plant and Machinery	500.00
Working Capital	50.00
Release of existing Working Capital	(15.00)
	535.00

II. Incremental Cash Inflow after Tax (CFAT)

(a) Generated by investment in India for 5 years

	$ Million
Sales Revenue (5 Million x $80)	400.00
Less: Costs	
Variable Cost (5 Million x $20)	100.00
Fixed Cost	30.00
Depreciation ($500Million/5)	100.00
EBIT	170.00
Taxes@35%	59.50
EAT	110.50
Add: Depreciation	100.00
CFAT (1-5 years)	210.50

(b) Cash flow at the end of the 5 years (Release of Working Capital) 35.00

(c) Cash generation by exports (Opportunity Cost)

	$ Million
Sales Revenue (1.5 Million x $80)	120.00
Less: Variable Cost (1.5 Million x $40)	60.00
Contribution before tax	60.00
Tax@35%	21.00
CFAT (1-5 years)	39.00

(d) Additional CFAT attributable to Foreign Investment

	$ Million
Through setting up subsidiary in India	210.50
Through Exports in India	39.00
CFAT (1-5 years)	171.50

III. Determination of NPV

Year	CFAT ($ Million)	PVF@12%	PV ($ Million)
1-5	171.50	3.6048	618.2232
5	35	0.5674	19.8590
			638.0822
Less: Initial Outflow			535.0000
			103.0822

Since NPV is positive the proposal should be accepted.

Practical QS

5. XYZ Ltd., a company based in India, manufactures very high quality modem furniture and sells to a small number of retail outlets in India and Nepal. It is facing tough competition. Recent studies on marketability of products have clearly indicated that the customers are now more interested in variety and choice rather than exclusivity and exceptional quality. Since the cost of quality wood in India is very high, the company is reviewing the proposal for import of woods in bulk from Nepalese supplier.

The estimate of net Indian (₹) and Nepalese Currency (NC) cash flows in Nominal terms for this proposal is shown below:

	Net Cash Flow (in millions)			
Year	0	1	2	3
NC	-25.000	2.600	3.800	4.100
Indian (₹)	0	2.869	4.200	4.600

The following information is relevant:

(i) XYZ Ltd. evaluates all investments by using a discount rate of 9% p.a. All Nepalese customers are invoiced in NC. NC cash flows are converted to Indian (₹) at the forward rate and discounted at the Indian rate.

(ii) Inflation rates in Nepal and India are expected to be 9% and 8% p.a. respectively. The current exchange rate is ₹ 1= NC 1.6

Assuming that you are the finance manager of XYZ Ltd., calculate the net present value (NPV) and modified internal rate of return (MIRR) of the proposal.

You may use following values with respect to discount factor for ₹ 1 @9%.

	Present Value	Future Value
Year 1	0.917	1.188
Year 2	0.842	1.090
Year 3	0.772	1

Solution — ICAI

5. Working Notes:

(i) Computation of Forward Rates

End of Year	NC	NC/₹
1	$NC1.60 \times \left(\frac{(1+0.09)}{(1+0.08)}\right)$	1.615
2	$NC1.615 \times \left(\frac{(1+0.09)}{(1+0.08)}\right)$	1.630
3	$NC1.630 \times \left(\frac{(1+0.09)}{(1+0.08)}\right)$	1.645

(ii) NC Cash Flows converted in Indian Rupees

Year	NC (Million)	Conversion Rate	₹ (Million)
0	-25.00	1.600	-15.625
1	2.60	1.615	1.61
2	3.80	1.630	2.33
3	4.10	1.645	2.49

Net Present Value

(₹ Million)

Year	Cash Flow in India	Cash Flow in Nepal	Total	PVF @ 9%	PV
0	---	-15.625	-15.625	1.000	-15.625
1	2.869	1.61	4.479	0.917	4.107
2	4.200	2.33	6.53	0.842	5.498
3	4.600	2.49	7.09	0.772	5.473
					-0.547

Modified Internal Rate of Return

	Year			
	0	1	2	3
Cash Flow (₹ Million)	-15.625	4.479	6.53	7.09
Year 1 Cash Inflow reinvested for 2 years (1.188 x 4.479)				5.32
Year 2 Cash Inflow reinvested for 1 years (1.090 x 6.53)				7.12
				19.53

$$MIRR = \sqrt[n]{\frac{\text{Terminal Cash Flow}}{\text{Initial Outlay}}} - 1 = \sqrt[3]{\frac{19.53}{15.625}} - 1 = 0.0772 \text{ say } 7.72\%$$

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